Tutorial Questions for CHM 111 for 2020/21 and 2021/22 academic session

1. How many molecules are there in 0.25 mol of glucose ($C_6H_{12}O_6$)?

- A) 1.505×10^{23}
- B) 3.011×10^{23}
- C) 4.516×10^{23}
- D) 6.022×10^{23}

2. What is the molar mass of calcium carbonate ($CaCO_3$)?

- A) 40.1 g/mol
- B) 60.1 g/mol
- C) 80.1 g/mol
- D) 100.1 g/mol

3. What is the name of the process in which an electric current is used to decompose a compound into its elements?

- A) Electrolysis
- B) Electroplating
- C) Electromagnetism
- D) Electrostatics

4. What are the electrodes made of in the electrolysis of molten sodium chloride?

- A) Copper
- B) Graphite
- C) Iron
- D) Platinum

5. What is the name of the principle that states that if a stress is applied to a system at equilibrium, the system will shift to relieve the stress?

- A) Le Chatelier's principle
- B) Boyle's law
- C) Dalton's law
- D) Hess's law

6. What will happen to the equilibrium position of the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, if the pressure is increased by decreasing the volume?

- A) The equilibrium will shift to the left, favoring the reactants.
- B) The equilibrium will shift to the right, favoring the products.
- C) The equilibrium will remain unchanged.
- D) The equilibrium will be disrupted and the reaction will stop.

7. What is the name of the peak that corresponds to the most abundant isotope in a mass spectrum?

- A) Base peak
- B) Molecular peak
- C) Parent peak
- D) Isotopic peak

8. What is the relative atomic mass of chlorine, if its natural abundance consists of 75.77% of chlorine-35 (mass = 34.969 u) and 24.23% of chlorine-37 (mass = 36.966 u)?

- A) 35.00 u
- B) 35.45 u
- C) 35.50 u
- D) 35.97 u

9. What is the effect of increasing the concentration of a reactant on a system at equilibrium?

- A) It shifts the equilibrium to the left, favoring the reactants.
- B) It shifts the equilibrium to the right, favoring the products.
- C) It has no effect on the equilibrium position.
- D) It depends on the temperature and pressure of the system.

10. What is the effect of decreasing the temperature on an exothermic reaction at equilibrium?

- A) It shifts the equilibrium to the left, favoring the reactants.
- B) It shifts the equilibrium to the right, favoring the products.
- C) It has no effect on the equilibrium position.
- D) It depends on the concentration and pressure of the system.

Tutorial 2

1. What is the Bronsted-Lowry definition of an acid and a base?

- A) An acid is a proton donor and a base is a proton acceptor.
- B) An acid is a proton acceptor and a base is a proton donor.
- C) An acid is an electron pair donor and a base is an electron pair acceptor.
- D) An acid is an electron pair acceptor and a base is an electron pair donor.

2. What are the conjugate acid and conjugate base of water in the following reaction: $H_2O + NH_3 \rightleftharpoons NH^{4+} + OH^-$?

- A) Conjugate acid: H_2O, conjugate base: OH^-
- B) Conjugate acid: NH^{4+}, conjugate base: NH_3
- C) Conjugate acid: H_2O, conjugate base: NH_3
- D) Conjugate acid: NH^{4+}, conjugate base: OH^-

3. What is the pH of a 0.01 M solution of hydrochloric acid (HCl)?

- A) 1
- B) 2
- C) 3
- D) 4

4. What is the pH of a 0.001 M solution of sodium hydroxide (NaOH)?

- A) 10
- B) 11
- C) 12
- D) 13

5. What is the pH of a buffer solution made by dissolving 0.15 mol of ammonia (NH_3, Kb = 1.8×10^{-5}) and 0.25 mol of ammonium chloride (NH_4Cl) in enough water to make 1 L of solution?

- A) 8.74
- B) 8.82
- C) 9.18
- D) 9.26

6. How much acetic acid (CH_3COOH, Ka = 1.8×10^{-5}) and sodium acetate (CH_3COONa) are needed to prepare 500 mL of a buffer solution with a pH of 4.8?

- A) CH_3COOH: 0.025 mol, CH_3COONa: 0.050 mol
- B) CH_3COOH: 0.050 mol, CH_3COONa: 0.025 mol
- C) CH_3COOH: 0.025 mol, CH_3COONa: 0.025 mol
- D) CH_3COOH: 0.050 mol, CH_3COONa: 0.050 mol

7. What is the equation that relates the pressure, volume, temperature and number of moles of an ideal gas?

- A) $PV = nRT$
- B) $PV = nRT^2$
- C) $PV = RT/n$
- D) $PV = n/RT$

8. What is the value of the universal gas constant R in SI units?

- A) 0.0821 L atm mol^{-1} K^{-1}
- B) 8.314 J mol^{-1} K^{-1}
- C) 62.4 L mmHg mol^{-1} K^{-1}
- D) 1.987 cal mol^{-1} K^{-1}

9. What is an exothermic reaction?

- A) A reaction that absorbs heat from the surroundings and has a positive enthalpy change.
- B) A reaction that absorbs heat from the surroundings and has a negative enthalpy change.
- C) A reaction that releases heat to the surroundings and has a positive enthalpy change.
- D) A reaction that releases heat to the surroundings and has a negative enthalpy change.

10. What is an endothermic reaction?

- A) A reaction that absorbs heat from the surroundings and has a positive enthalpy change.
- B) A reaction that absorbs heat from the surroundings and has a negative enthalpy change.
- C) A reaction that releases heat to the surroundings and has a positive enthalpy change.
- D) A reaction that releases heat to the surroundings and has a negative enthalpy change.

Tutorial 3

1. What is the standard enthalpy change of reaction?

- A) The enthalpy change when a reaction occurs in the molar quantities shown in the chemical equation under standard conditions.
- B) The enthalpy change when a reaction occurs in the molar quantities shown in the chemical equation under any conditions.
- C) The enthalpy change when a reaction occurs in any quantities under standard conditions.
- D) The enthalpy change when a reaction occurs in any quantities under any conditions.

2. What are standard conditions for thermochemistry?

- A) 25°C and 1 atm pressure for all substances.
- B) 0°C and 1 atm pressure for all substances.
- C) 25°C and 1 bar pressure for all substances.
- D) 0°C and 1 bar pressure for all substances.

3. What mass of copper will be deposited at the cathode from the electrolysis of aqueous copper sulfate using a current of 0.50 A for 30 minutes? (Relative atomic mass of Cu = 63.5, Faraday constant = 96500 C mol^{-1})

- A) 0.197 g
- B) 0.394 g
- C) 0.591 g
- D) 0.788 g

4. What is the volume, in dm^3, of 0.25 mol of oxygen gas at room temperature and pressure? (Molar volume of gas at r.t.p. = 24.0 dm^3 mol^{-1})

- A) 6.0 dm^3
- B) 12.0 dm^3
- C) 48.0 dm^3
- D) 96.0 dm^3

5. What is the Ka value of a weak acid that has a pH of 4.5 when its concentration is 0.1 M? (Ka = $[H^+][A^-]/[HA]$, where HA is the weak acid and A^- is its conjugate base)

- A) 1×10^{-6}
- B) 2×10^{-6}
- C) 3×10^{-6}

- D) 4×10^{-6}

6. What is the hydroxide ion concentration of a solution that has a pOH of 4?

- A) 1×10^{-4} M
- B) 4×10^{-4} M
- C) 1×10^{-10} M
- D) 4×10^{-10} M

7. How much acetic acid (CH_3COOH, Ka = 1.8×10^{-5}) and sodium acetate (CH_3COONa) are needed to prepare 500 mL of a buffer solution with a pH of 4.8? (Use the Henderson-Hasselbalch equation: pH = pKa + log([base]/[acid]))

- A) CH_3COOH: 0.025 mol, CH_3COONa: 0.050 mol
- B) CH_3COOH: 0.050 mol, CH_3COONa: 0.025 mol
- C) CH_3COOH: 0.025 mol, CH_3COONa: 0.025 mol
- D) CH_3COOH: 0.050 mol, CH_3COONa: 0.050 mol

8. What is the pressure exerted by 0.5 mol of nitrogen gas in a container of volume 10 L at a temperature of 300 K? (Gas constant R = 8.314 J/mol K)

- A) 0.125 atm
- B) 0.250 atm
- C) 1.000 atm
- D) 2.000 atm

9. What is the enthalpy change when 50 g of ice at -10°C is melted to water at 25°C? (Specific heat capacity of ice = 2.11 J/g °C, specific heat capacity of water = 4.18 J/g °C, enthalpy of fusion of ice = 334 J/g)

- A) -16700 J
- B) -10550 J
- C) +10550 J
- D) +16700 J

10. What is the standard enthalpy change of reaction for the following equation:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l),$$

given the following enthalpies of formation: $H_2(g)$ = 0 kJ/mol, $O_2(g)$ = 0 kJ/mol, $H_2O(l)$ = -285.8 kJ/mol? (Use Hess's law: $\Delta Hrxn = \Sigma\Delta Hf(products) - \Sigma\Delta Hf(reactants)$)

- A) +571.6 kJ/mol
- B) -571.6 kJ/mol
- C) +285.8 kJ/mol
- D) -285.8 kJ/mol

Tutorial 4

1. What is the rate law for the following reaction: 2A + B → C + D, given that the rate doubles when the concentration of A doubles and quadruples when the concentration of B doubles?

- A) rate = k[A][B]
- B) rate = $k[A]^2[B]$
- C) rate = $k[A][B]^2$
- D) rate = $k[A]^2[B]^2$

2. What is the activation energy for a reaction that has a rate constant of 4.82×10^{-4} s^{-1} at 298 K and a frequency factor of 6.22×10^{13} s^{-1}? (Gas constant R = 8.314 J/mol K)

- A) 50.7 kJ/mol
- B) 75.5 kJ/mol
- C) 100.9 kJ/mol
- D) 151.0 kJ/mol

3. What is the order of a reaction that has a half-life that is independent of the initial concentration of reactants?

- A) 0
- B) 1
- C) 2
- D) Cannot be determined

4. What are the units of the rate constant k for a second-order reaction?

- A) M/s
- B) s^{-1}
- C) M^{-1} s^{-1}
- D) M^{-2} s^{-1}

5. How does the value of the rate constant k change with temperature for an endothermic reaction?

- A) It increases as temperature increases.
- B) It decreases as temperature increases.
- C) It remains constant as temperature changes.
- D) It depends on the activation energy of the reaction.

6. How does adding a catalyst affect the rate of a chemical reaction?

- A) It increases the rate by lowering the activation energy and increasing the number of effective collisions.
- B) It increases the rate by raising the activation energy and increasing the number of effective collisions.
- C) It decreases the rate by lowering the activation energy and decreasing the number of effective collisions.
- D) It decreases the rate by raising the activation energy and decreasing the number of effective collisions.

7. What is the molecularity of an elementary process?

- A) The number of molecules involved in the process.
- B) The number of moles involved in the process.
- C) The number of atoms involved in the process.
- D) The number of electrons involved in the process.

8. What is the rate law for a termolecular elementary process involving one molecule of A, one molecule of B, and one molecule of C?

- A) rate = k[A]
- B) rate = k[A][B]
- C) rate = k[A][B][C]
- D) rate = $k[A]^2[B]$

9. What is the integrated rate law for a first-order reaction?

- A) [A] = -kt + [A]0
- B) ln[A] = -kt + ln[A]0

- C) 1/[A] = kt + 1/[A]0
- D) None of the above

Tutorial 5

1. What is the order of a reaction that has a rate law of the form rate = $k[A]^2[B]^2$?
   - A) 0
   - B) 1
   - C) 2
   - D) 4

3. What are the units of the rate constant k for a second-order reaction?
   - A) M/s
   - B) s^{-1}
   - C) $M^{-1}\ s^{-1}$
   - D) $M^{-2}\ s^{-1}$

3. What is the integrated rate law for a second-order reaction?

- A) [A] = -kt + [A]0
- B) ln[A] = -kt + ln[A]0
- C) 1/[A] = kt + 1/[A]0
- D) None of the above

4. What is the half-life of a second-order reaction with a rate constant of 0.01 $M^{-1}\ s^{-1}$ and an initial concentration of 0.1 M?

- A) 5 s
- B) 10 s
- C) 20 s
- D) Depends on the initial concentration

5. What is the first law of thermodynamics?
   - A) The internal energy of an isolated system is constant.
   - B) The internal energy of an isolated system increases over time.
   - C) The internal energy of an isolated system decreases over time.

- D) The internal energy of an isolated system is zero.

6. What is an example of a strong acid that can conduct electricity well when dissolved in water?

    - A) Hydrochloric acid (HCl)

    - B) Acetic acid (CH_3COOH)

- C) Carbonic acid (H_2CO_3)

- D) Citric acid ($C_6H_8O_7$)

7. What is an example of a weak acid that can conduct electricity poorly when dissolved in water?

    - A) Nitric acid (HNO_3)

    - B) Sulfuric acid (H_2SO_4)

    - C) Phosphoric acid (H_3PO_4)

    - D) Hydroiodic acid (HI)